

Module 02: Agents — Each Maker at the Table

Learning Objectives

1. Recognize each circle member as an autonomous inference agent with their own generative model — shaped by their unique skill level, project history, and craft preferences.
2. Describe how the diversity of generative models within a circle creates conditions for collective learning and mutual enrichment.
3. Analyze the crochet circle as a collective agent whose group-level behavior emerges from the interactions of individual agents.

Introduction

Look around your crochet circle. Each person at the table is working on something different — a baby blanket in pastel cotton, a lacy shawl in mohair, a simple granny square practice piece. Each person holds their hook differently, tensions their yarn in their own way, and carries a unique history of projects completed, patterns attempted, and stitches mastered. Each person, in the language of Active Inference, is an **agent** — an autonomous system that perceives, decides, and acts according to its own internal model of the world.

In Module 1, we saw the circle as a system. Now we zoom in to the individuals who compose it. Understanding each maker as an agent — with their own **generative model**, their own **prediction errors**, and their own **free energy** to minimize — is essential to understanding how circles function, why they are so valuable, and how collective intelligence emerges from the interplay of diverse minds and hands.

Key Concepts

1. The Generative Model of a Crocheter

Every agent in Active Inference carries a **generative model** — an internal representation of how the world works, which the agent uses to predict what will happen next and to guide its actions. For a crocheter in a circle, this generative model includes:

- **Technical knowledge:** Which stitches they know, how patterns are structured, what different hook sizes and yarn weights do. A beginner's generative model might include chain, single crochet, and double crochet. An experienced maker's model includes cables, colorwork, Tunisian techniques, freeform sculpture, and an intuitive sense of how different fibers behave.
- **Project expectations:** What the finished piece should look like, how many rows remain, what the gauge should be. These are predictions about the future state of the work.
- **Social knowledge:** Who in the circle is good at what, who to ask about a particular technique, what topics of conversation are welcome, how the group dynamics flow.
- **Aesthetic preferences:** What colors, textures, and styles appeal to them. These are prior beliefs that shape which projects they choose and how they evaluate their own work.

The richness of a person's generative model is directly related to their experience. A beginner has a **shallow model** — few predictions available, high uncertainty about outcomes. They pick up the hook and are genuinely unsure whether the next stitch will produce the intended result. An expert has a **deep model** — layers of experience that allow precise predictions about what a stitch will do, what a yarn will feel like after washing, how a pattern will drape on a body.

2. Diversity of Models in the Circle

One of the most powerful features of a crochet circle is that it brings together agents with **different generative models**. This diversity is not a bug; it is the engine of collective intelligence.

Consider a circle with five members: - **Rosa** has crocheted for forty years. Her generative model is deep and wide — she can look at a finished garment and mentally reverse-engineer

the pattern. She has strong priors about how things "should" be done. - **James** learned to crochet two months ago. His model is narrow but growing fast. Every meeting brings prediction errors that drive rapid learning. - **Priya** is an adventurous intermediate. She tries techniques that are slightly beyond her current model, deliberately seeking prediction error as a learning strategy. - **Devon** is an expert in colorwork but has never done amigurumi. Their model is deep in one area and shallow in another. - **Mei** comes from a tradition of Chinese crochet lace and carries a generative model shaped by a different cultural lineage than the Western patterns most of the group uses.

When these agents interact, something remarkable happens. Rosa's deep model can reduce James's prediction error by explaining why his stitches are uneven. Mei's different cultural model introduces the group to techniques that expand everyone's generative model. Priya's adventurous spirit exposes the group to new patterns that the more cautious members would never have tried alone. Devon's depth in colorwork benefits those working on multi-color projects, while their beginnership in amigurumi lets others experience the pleasure of teaching.

The circle, in this sense, functions as a **model-expansion machine**. Each agent arrives with their own generative model; through interaction, all models grow.

3. The Circle as a Collective Agent

Beyond the individual agents, the circle itself can be understood as a **collective agent** — a higher-order entity that perceives, decides, and acts in ways that no individual member does alone.

The collective agent has its own generative model: the group's shared understanding of what projects are feasible, what difficulty level is appropriate for a group challenge, what the local craft community values. It has its own preferences: the circle might gravitate toward practical items (blankets, hats) rather than decorative ones, or vice versa. It has its own active inference: when the circle decides to enter a craft fair, it is the group — not any one member — making a prediction about the future (we will have enough finished items, the fair will be worthwhile) and acting to make that prediction come true.

The emergence of the collective agent from individual agents is not magic — it arises through the mechanisms we will explore in later modules (communication, shared perception,

coordinated action). But it is real. Ask any long-time member of a crochet circle, and they will tell you: the group has a personality, a style, a way of being that is distinct from any of its members taken individually.

4. Autonomy Within Community

A crucial feature of agents in Active Inference is **autonomy** — each agent acts according to its own generative model, minimizing its own free energy. In a crochet circle, this means each person works on their own project, makes their own decisions about stitches and colors, and maintains their own creative vision.

This autonomy is not threatened by community; it is enriched by it. The circle does not demand uniformity. Rosa does not have to crochet like James, and James does not have to crochet like Rosa. Each agent maintains its own boundary (its own Markov blanket) while participating in the larger system.

The moments when autonomy and community intersect are some of the most generative in a circle's life. When Priya asks Rosa for help with a complex stitch, she is not giving up her autonomy — she is using the social environment to update her own generative model. When the circle decides on a group project, each member interprets and executes the project according to their own model, producing a collection of works that are related but not identical.

Applications

- **Agent Heterogeneity as Resource:** In many social settings, differences in skill level are seen as problems to manage. In a crochet circle understood through Active Inference, heterogeneity is a resource. The beginner's fresh eyes and frequent prediction errors keep the group attentive to fundamentals. The expert's deep model provides stability and reduces collective uncertainty. The adventurous intermediate drives exploration. A circle composed entirely of experts or entirely of beginners would be a less effective learning environment than one with a healthy mix.
- **The Quiet Agent:** Not every agent in a circle is verbally active. Some people crochet silently, absorbing the conversation and the visual information from others' work. In

Active Inference, these quiet agents are still perceiving, predicting, and learning. Their generative models are updating even when they are not speaking. Recognizing this helps circles value all forms of participation, not just the vocal ones.

Conclusion

Every person at the crochet circle table is an autonomous inference agent, carrying a unique generative model shaped by experience, culture, and preference. The circle thrives because of the diversity of these models — beginners and experts, adventurers and traditionalists, talkers and quiet observers all contribute to a collective intelligence that exceeds what any individual could achieve alone. The circle itself becomes a collective agent, with emergent properties that arise from the interaction of its members. In the next module, we explore how these agents perceive one another — how they read the room while they stitch.